

Chasing the Nines

Reliable RL Across States and Seeds



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Pendulum: 2,501 starts × five seeds **Supervision reduces the reliability tail from 497 failures to nine. The pure-RL critics can repair many actions that the raw actor gets wrong.**

01 Measure the tail, not only the mean

RL + supervised	Pure RL
99.928%	96.026%
9 / 12,505 failures	497 / 12,505 failures
Task 93.858%	Task 92.875%
Strict wins 12.555%	Strict wins 22.823%

Near reference: within 5 return points. **Task:** upright and slow for 80%. **Strict win:** higher return than the reference.

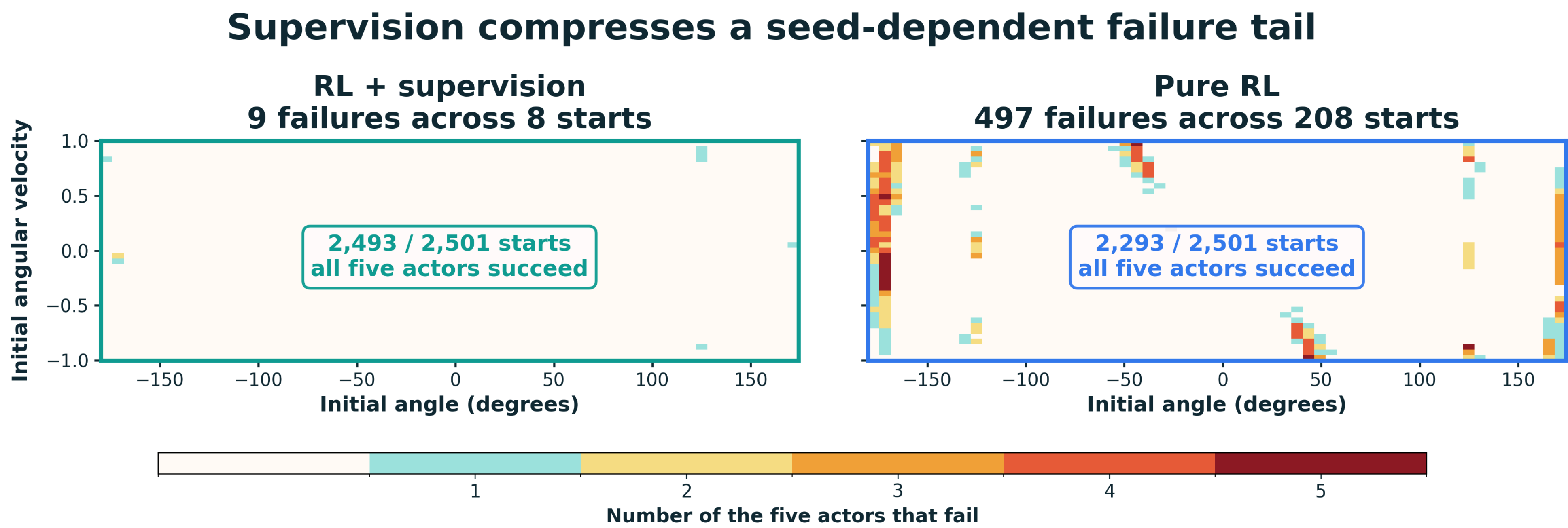
RL + supervised
Dagger labels the states that the learner actually visits. Automatic follow-up revisits measured weak starts.

30k learner-state labels	20k weak-start follow-up	50k critic + local Q- search
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Pure RL
FastSACN8 trains categorical critics on one-step and eight-step returns, with two critic updates per transition.

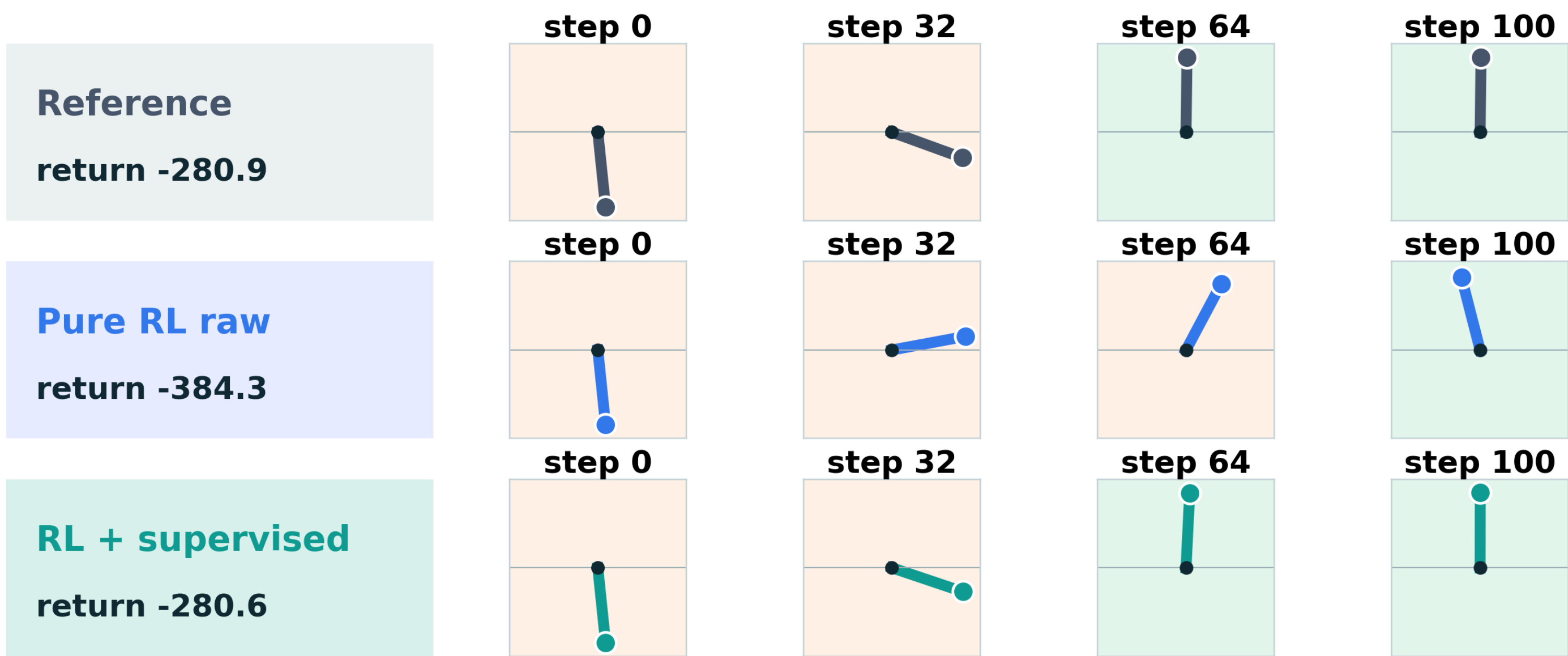
100k reward transitions	Symmetry physical reflection	Search 41 torques, twin- Q agreement
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02 Supervision removes most failure regions



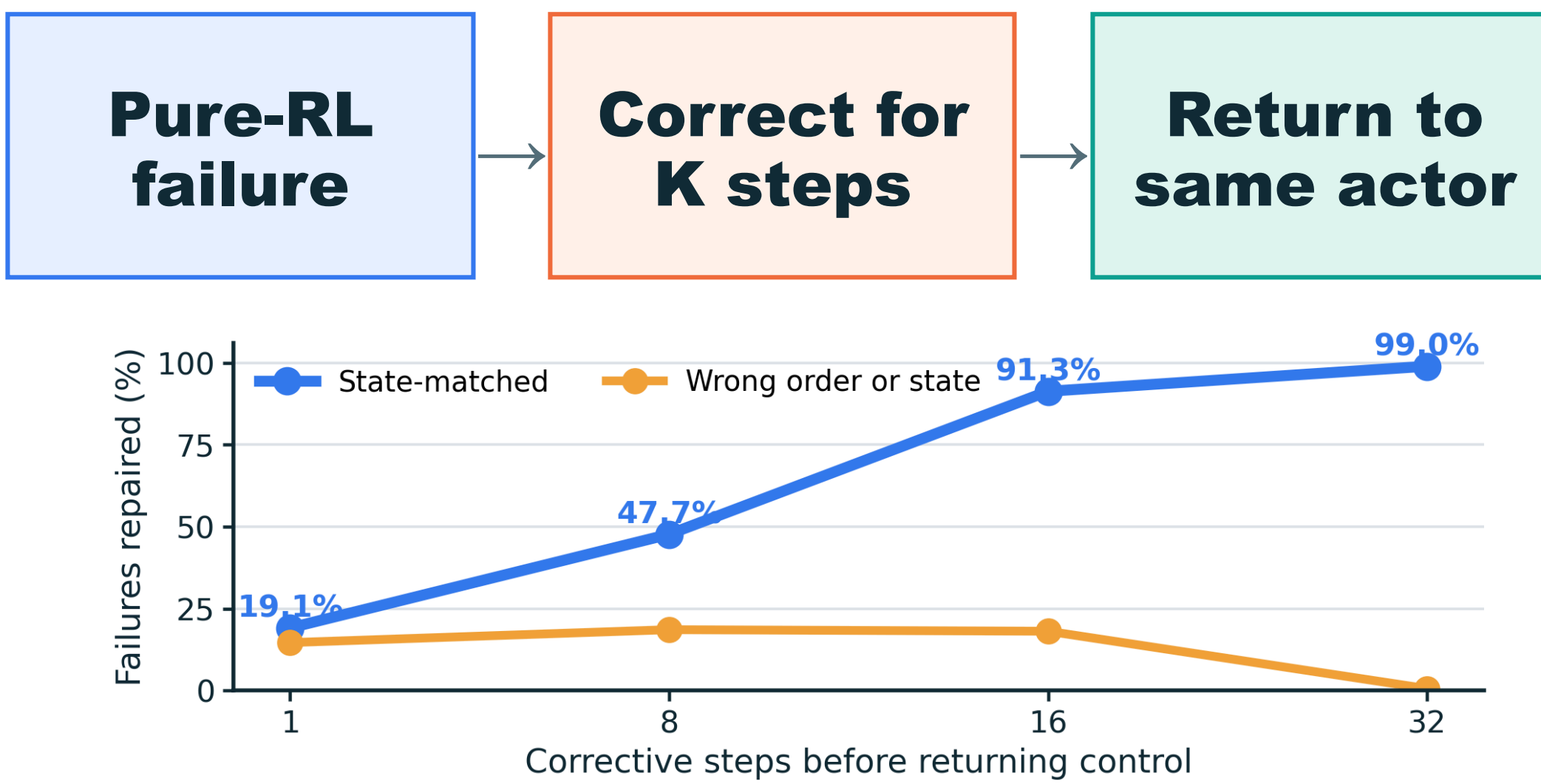
All five mixed actors succeed on 2,493 of 2,501 starts. For pure RL, all five succeed on 2,293, outcomes vary by seed on 195, and all five fail on 13.

03 The gap appears during the closed-loop recovery



From the same hard start, the reference and mixed actor are upright by step 64. The raw pure-RL actor is still drifting and finishes 103.7 return points lower.

04 A failure usually spans several decisions



Among 1,267 raw-actor failures, one step repairs 19.1%; 16 state-matched steps repair 91.3%. Shuffling the actions across states destroys the effect.

05 What the diagnostics reveal

+14 points Critic-guided action changes help more after multi-step critic training.	+770 repairs Reflection and twin-Q search repair 770 seed-state trials.
1,010 / 1,021 Failures have a reference-like action in nearby replay.	70.3% at limit Selected raw versus one-step SAC: 70.3% versus 59.1%, five paired seeds.

Broad missing coverage and dead features do not explain the gap. The raw actor is saturated more often, and correcting one action usually does not stop later drift.

Mechanistic story: The critics often identify a better immediate action. Supervision also teaches the actor what to do after that action changes the next state.